

EMERGENCY VEHICLE DETECTION AND TRIGGERING SIGNAL

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Project Final Thesis

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Declaration page of the candidates & supervisor

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Signature of the supervisor:

Date.....

(Prof Nuwan Kodagoda)

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Signature of the Co-supervisor:

Date.....

(Ms. Karthiga Rajendran)

Abstract

Urban traffic congestion significantly impacts the response time of emergency services such as ambulances, fire and rescue units, and police vehicles. Due to their fixed timing mechanisms and inability to dynamically adjust to emergency conditions, traditional traffic signal systems frequently result in significant delays. To improve traffic flow and shorten reaction times, this study offers a reliable multi-modal emergency vehicle identification and traffic signal pre-emption system linked with the Simulation of Urban Mobility (SUMO) platform.

For accurate detection, the developed system integrates computer vision and sound analysis. Emergency cars are visually identified in real time using a deep learning model based on YOLOv8. Simultaneously, an audio detection module uses the YAMNet model and log-mel spectrograms to categorize siren noises. Although acoustic sensors offer a non-line-of-sight benefit, low signal-to-noise ratios in urban settings frequently cause false detections due to background noise.

To improve robustness, a multi-stage verification process is introduced, incorporating HSV-based flashing light detection and OCR-based text recognition of keywords such as “AMBULANCE,” “POLICE,” and “FIRE & RESCUE.” Additionally, a top-view detection approach is used to identify emergency vehicles even under occlusion by large vehicles.

According to experimental data, the system maintains real-time performance while achieving over 82% detection accuracy with an overall mAP@0.5 of 0.774. To improve the effectiveness of emergency response Upon detection, the system leverages Multi-Agent Reinforcement Learning (MARL) to coordinate between intersections and automatically creates green corridors by interacting with the SUMO environment via the TraCI interface.

Keywords - Emergency Vehicle Detection, YOLOv8, Multi-Modal Fusion, Acoustic Detection, OCR, HSV Colour Segmentation, Multi-Agent Reinforcement Learning, Simulation of Urban Mobility(SUMO), Traffic Control Interface (TraCI), Smart Traffic Systems.

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List Of Abbreviations

SUMO	Simulation of Urban Mobility
YOLO	You Only Look Once
HSV	Hue, Saturation, Value
TraCI	Traffic Control Interface
MARL	Multi-Agent Reinforcement Learning
ML	Machine Learning
NN	Neural Network
MFCC	Mel-Frequency Cepstral Coefficients
OCR	Optical Character Recognition

Table 1 : List of Abbreviations

1. Introduction

1.1. Background Study and Literature Review

1.1.1. Background Study

Urban traffic congestion is a pervasive issue that severely hampers the efficacy of emergency services. In critical scenarios, seconds can mean the difference between life and death. Traditional traffic signal systems operate on fixed-time cycles or basic sensor inputs that lack the intelligence to adapt to dynamic emergency scenarios. Consequently, emergency vehicles (EVs) often face significant delays at busy intersections, increasing response times and risking public safety [2].

Traffic congestion can have life-threatening consequences, particularly for patients requiring urgent medical attention. In densely populated urban areas, emergency ambulances often experience significant delays due to blocked or poorly managed traffic conditions. A reported incident in The Times of India highlights that several critically injured individuals lost their lives while being transported to hospitals, primarily because ambulances were stuck in severe traffic jams [1]. These delays reduce the chances of timely medical intervention, directly impacting patient survival rates. Figure 2 shows that when traffic flow is not efficiently controlled, emergency vehicles face extended waiting times at intersections, leading to fatal outcomes. This emphasizes the importance of intelligent traffic management systems capable of prioritizing emergency vehicles to ensure faster response and safer transportation.



Figure 1: Ambulance Stucks in Traffic Road



Figure 2: Ambulance in the middle of the traffic jam

The recognition of emergency vehicles is a crucial part of intelligent transportation systems (ITS) that are intended to increase traffic safety and speed up police, fire, and ambulance response times. The primary goal of emergency vehicle detection is to automatically recognize their presence in traffic and initiate priority actions, such as lane clearance, traffic signal pre-emption, or warning drivers and vehicles in the vicinity.

In the past, infrastructure-based and manual methods, like radio communication between traffic control centres and emergency services, were used to detect emergency vehicles. Nevertheless, these techniques are constrained by their lack of scalability in crowded metropolitan settings, latency, and reliance on human involvement.

Modern methods mostly rely on sensor-based and AI-driven methods due to technological advancements. One of the most popular techniques is computer vision-based detection, in which real-time video streams are captured by cameras placed at junctions or on roadside infrastructure. Using visual characteristics like siren lights, vehicle shape, colour patterns, and markings like text or labels in the emergency vehicles, deep learning models like Convolutional Neural Networks (CNNs) and object detection algorithms like YOLO (You Only Look Once) are trained to identify emergency vehicles. These models are appropriate for smart traffic systems because of their great accuracy and capacity for real-time processing.

Once detected, the system can send real-time SMS alerts to nearby traffic police for immediate action. At the same time, the signal control system can adjust traffic lights

(green wave) to give priority passage to the emergency vehicle. This combined approach helps reduce congestion and ensures faster and safer movement of emergency vehicles.

1.1.2. Literature Review

Emergency vehicle detection is a vital component in modern intelligent transportation systems, particularly within the context of smart cities. The increasing volume of urban traffic often leads to significant delays for ambulances, fire trucks, and police vehicles, especially at busy intersections with fixed-time traffic signals. These delays can have life-threatening consequences, as emergency response time is a critical factor in saving lives and property. [2]

Current solutions for Emergency Vehicle detection largely rely on single-modality approaches Figure 3 shows that , such as visual recognition using Convolutional Neural Networks (CNNs) or simple siren detection methods.



Figure 3:Single modality Siren Sound Detection

However, these methods have inherent limitations. Visual-only systems struggle in low-light conditions, adverse weather (fog or heavy rain), or when the line of sight is obstructed by larger vehicles [3]. Conversely, acoustic-based siren detection is vulnerable to urban noise pollution, such as construction sounds, horns, and engine noise, leading to high false-positive rates [7].

To address these challenges, this research proposes a Multi-Modal Emergency Vehicle Detection and Triggering System. The core novelty lies in the fusion of three distinct detection modalities: (1) Acoustic detection using YAMNet for early warning, (2) Visual detection using YOLOv8 for vehicle classification, and (3) Logical verification via HSV color masking and Optical Character Recognition (OCR). This redundant architecture

ensures high reliability under diverse environmental conditions. The system is integrated into the Simulation of Urban Mobility (SUMO) platform, enabling real-time traffic signal pre-emption through the Traffic Control Interface (TraCI). Beyond simulation, the solution extends to physical hardware implementation, controlling traffic lights and barrier gates via an Arduino microcontroller. This paper details the system architecture, methodology, and expected outcomes of the Traffixion system.

1.2. Research Gap

Traditional method of emergency vehicle detection and prioritize them was human based, it requires human energy and cooperative support of other drivers. Higher chance of accidents when crossing red lights or navigating traffic. In the other hand, Upcoming Vehicle Detection (VD) systems rely on vehicle-mounted sensors for Vehicle-to-Vehicle (V2V) communication, requiring every vehicle to be equipped with compatible hardware. These systems operate independently from traffic signals, offering no direct means to reduce waiting times. To satisfy both parties with equal requirements smart based vehicle detection and prioritizing in an interaction way using MARL is the main the planned proposal. The proposed Emergency Vehicle Detection component overcomes these limitations by using multi-modal models' visual recognition, siren sound analysis, and HSV colour segmentation to identify emergency vehicles in real time. Integrated with SUMO for simulation and a dashboard for visualization, it enables traffic signal pre-emption, showcasing dynamic green corridors to authorities for improved emergency response.

Below is a concise gap analysis comparing the system (supervised image classification + HSV light detection + siren audio detection) with existing methods. The analysis highlights limitations of current approaches and novelty contributions of your integrated solution, with comparison of similar research topics with their relevant references.

Table 2 shows the comparison of the technical components that were used in some research papers, every individual component uses to identify specify details using ML models. However, these components encounter failures at certain stages. To overcome these limitations and achieve improved performance, the proposed system adopts a multi-component, microservices-based architecture.

Method	Strengths	Research Gaps	References
Pure Image Classification	High accuracy in controlled environments; real-time inference with lightweight CNNs	✗ Fails in low visibility (fog/night). ✗ Struggles with obscured emergency ✗ Limited to visual range only	[3] [4]
HSV Based Light Detection	Robust for colour-specific detection (red/blue lights); low computational cos	✗ High false positives (e.g., brake lights, traffic signals). ✗ Sensitive to environmental lighting. ✗ Ignores non-visual cues (sirens).	[5] [6]
Siren Sound Detection	Effective for non-line-of-sight scenarios; leverages acoustic uniqueness.	✗ Vulnerable to urban noise pollution (horns, engines). ✗ Limited range. ✗ Cannot classify vehicle type visually.	[7] [8]
Sound + Light Detection	Reduces false negatives via multi-modal data.	✗ Often excludes explicit light-feature extraction. ✗ Underutilizes temporal flashing patterns of lights. ✗ Rarely optimized for emergency vehicles specifically.	[9] [10] [11]

Table 2 : Competitive Analysis

1.3. Research Problem

In modern urban environments, emergency response effectiveness is heavily influenced by the ability of traffic management systems to detect emergency vehicles in real-time and grant them priority passage. Traditional traffic signal systems operate on fixed timers or basic sensor inputs and lack the capability to dynamically adapt for emergency situations. While some systems incorporate visual detection, they often fail under poor visibility, occlusion, or adverse weather conditions. Similarly, audio-based siren detection can be unreliable in noisy urban settings or when the signal-to-noise ratio is low.

Existing approaches tend to rely on a single detection modality either visual, audio, or light-based recognition which limits robustness and accuracy. Moreover, these detection systems are often tested in isolation and not integrated into realistic urban traffic simulations to evaluate their impact on reducing response times. The absence of multi-modal fusion combining computer vision, acoustic recognition, and hue-based light bar segmentation further reduces the adaptability of current solutions.

Additionally, current research rarely integrates real-time detection with a traffic control simulation framework like SUMO to dynamically trigger and optimize signal phases for emergency vehicle clearance. This gap prevents stakeholders from accurately assessing the operational benefits, scalability, and constraints of such systems in complex traffic networks. [11] [12] [13] [14]

The proposed methodology focuses on developing a multi-modal emergency vehicle detection and traffic signal pre-emption system, integrating visual, acoustic, and hue-based detection modules with the SUMO simulation environment through the TraCI API. The approach is divided into key phases as follows:

1.4. Research Objectives

1.4.1. Main Objectives

This research aims to design and develop a multi-modal emergency vehicle detection system within the SUMO traffic simulator. To accurately identify emergency vehicles caught in urban traffic blockages, the system integrates frontal and top-view visual data utilizing YOLO and OpenCV, Optical Character Recognition (OCR) for vehicle label extraction, acoustic siren

detection, and hue-based light bar recognition. Upon real-time detection, the system dynamically triggers traffic signal pre-emption to establish a clear corridor, ultimately reducing emergency response times in congested urban environments.

1.4.2. Specific Objectives

- **Visual Detection & OCR Integration:** A YOLO and OpenCV-based real-time module was developed to identify emergency vehicles from frontal and top-view traffic camera feeds under varying lighting and weather conditions. Optical Character Recognition (OCR) was integrated to extract specific vehicle labels.
- **Acoustic Siren Recognition:** A machine learning model was built and trained to detect and classify emergency sirens amidst the ambient noise of complex urban environments.
- **Hue-Based Light Bar Detection:** HSV color segmentation combined with Haar Cascade classifiers was implemented to accurately detect emergency light bars, thereby improving overall detection reliability.
- **SUMO & TraCI Integration:** The detection system was connected to the SUMO traffic simulator via TraCI, which enabled the dynamic modification of traffic signal phases in real-time for emergency vehicle prioritization.
- **System Performance Evaluation:** The system's detection accuracy, processing latency, and overall impact on emergency vehicle travel times were assessed through simulation experiments across various traffic scenarios.
- **Instant notification to the relevant parties :** If emergency vehicles are detected the details will be sent via text messages to the traffic authorities.
- **Dashboard Visualization:** A comprehensive dashboard was designed to display real-time detection metrics to end-users, and a system was established to automatically relay critical data to relevant authorities.

1.4.3. Business Objectives

Primary Business Objective

Reduce Emergency Response Times (ERT): To significantly decrease the time it takes for emergency vehicles to navigate through congested urban traffic, directly contributing to faster medical interventions, reduced property damage, and improved survival rates.

Operational Efficiency Objectives

Automate Traffic Signal Pre-emption: To eliminate the reliance on manual traffic control center interventions by fully automating the clearance of intersections, allowing traffic authorities to reallocate human resources to other critical tasks.

Optimize City-Wide Traffic Flow: To minimize the secondary congestion caused by emergency vehicles by predicting their routes (via indicator detection) and proactively updating neighboring traffic agents, effectively rerouting civilian traffic before a bottleneck occurs.

System Reliability & Risk Management Objectives

Ensure All-Weather Operational Reliability: To provide a detection system that maintains high accuracy regardless of environmental constraints (e.g., low light, rain, glare, and high ambient city noise), thereby reducing the risk of system failure during critical emergencies.

Minimize False Alarms and Traffic Disruption: To implement a multi-modal fusion approach (Visual + Acoustic + Light Bar) that verifies an emergency before triggering signal pre-emption, preventing unnecessary disruptions to standard traffic flows.

Stakeholder Empowerment Objectives

Provide Real-Time Situational Awareness: To equip traffic management authorities and first responders with a centralized dashboard that offers live visibility into emergency vehicle locations and system actions.

Facilitate Data-Driven Decision Making: To establish a system that logs detection accuracy, latency, and travel-time metrics, providing city planners with concrete data to evaluate and improve future urban traffic infrastructure.

Strategic Scalability Objectives

Validate a Scalable Smart-City Framework: To successfully prove the integration of advanced AI (YOLO, OCR, ML) with existing traffic simulation software (SUMO/TraCI), demonstrating a cost-effective, scalable blueprint for future real-world Smart City deployments.

2. METHODOLOGY

2.1. Methodology

The methodology served as the foundation of this applied research, laying out the systematic processes, instruments, and strategies used to gather, process, and interpret data in accordance with the project's objectives. Particularly in projects involving computer vision, acoustic signal processing, and real-time traffic simulation, a clearly defined methodology guaranteed that the development process was transparent, reproducible, and grounded in sound logic. To address the intricate and iterative requirements of building a multi-modal emergency vehicle detection system integrated with the SUMO environment, this study employed a hybrid methodology that combined a seven-stage System Development Life Cycle (SDLC) with Agile software development practices.

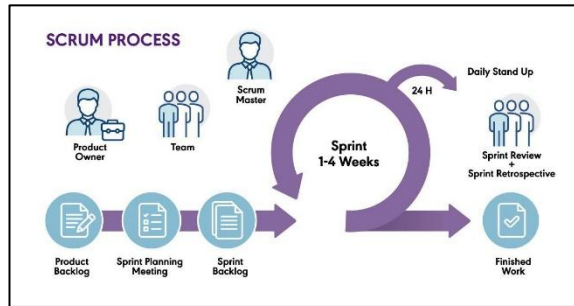


Figure 4: Agile Scrum Framework

Agile Methodology for Research Development: The main project management framework for this study was Agile, a methodology originally created for software development but increasingly popular in multidisciplinary research due to its adaptability, iterative structure, and evolution driven by stakeholder feedback. Agile was the perfect choice for a dynamic project with changing traffic management requirements and evolving technical complexities, as it placed a strong emphasis on collaboration, modularity, and continuous testing. Core modules like YOLO-based visual detection, OCR label extraction, acoustic siren recognition, and hue-based light bar detection were able to be seamlessly integrated into the SUMO simulator thanks to the Agile model.

In particular, the Scrum version of Agile was used, in which the work was broken down into smaller units called sprints, which were usually two to three weeks long. A major system component (such as multi-angle visual processing, acoustic model training, or TraCI signal pre-emption logic) was the focus of each sprint. Continuous improvement and prompt issue resolution were made possible by regular sprint reviews, daily stand-up meetings, and

retrospectives. The Scrum cycle used in this study is depicted in Figure 5, which shows how tasks moved through iterative feedback loops from planning to delivery.

Seven-Stage Development Framework: Based on the ideas of Agile methodology, the research process developed over the course of a structured seven-stage process. Requirements collection, system design, deployment, testing, maintenance, and continuous improvement were all phases in this process. The results and input from the previous stage were used to inform each phase, resulting in a dynamic and adaptable development loop. The team was able to address issues as they arose, improve technical elements, and adjust to changing system needs thanks to this iterative structure. Agile's focus on adaptability was especially helpful in handling the project's multidisciplinary elements, which integrated cutting-edge computer vision and acoustic machine learning methods with real-time traffic simulation constraints. The development cycle encouraged methodical advancement while providing room for creativity, user input, and iterative improvement. Figure 6 shows the seven-stage development life cycle, and the rest of the methodology section will explain each of the phases of the life cycle.

Iterative Development and Collaboration: Cross-functional cooperation and incremental development cycles were key factors in the multi-modal detection system's success. To evaluate module performance and adjust to evolving needs, the interdisciplinary team—which included software developers, computer vision engineers, and traffic simulation experts—convened biweekly review sessions. Agile collaboration tools like Jira and GitHub Projects made it easier to track issues, documentation, and code changes in real time. This iterative approach proved particularly helpful in improving YOLO detection accuracy under varying lighting conditions, filtering urban noise from the acoustic siren detection, and fine-tuning the TraCI connection with the SUMO environment.

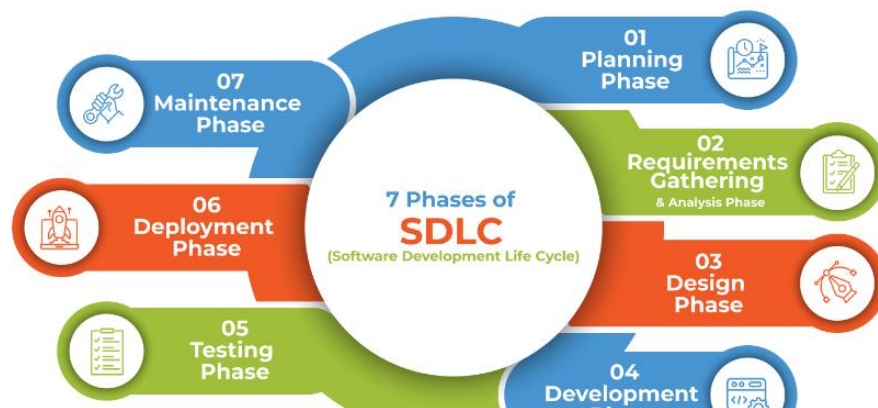


Figure 5 : Seven Stage of Software Development Life Cycle

Flexibility and Responsiveness: Agile methodology's adaptability and responsiveness were essential in overcoming the ever-changing obstacles that arose during the detection system's development. Unexpected problems like latency in real-time video processing, false positives in HSV light bar detection, and synchronization delays between the detection modules and the SUMO simulator surfaced as the project progressed.

Because Agile is iterative, the team was able to quickly adjust, optimizing OpenCV pipelines, refining acoustic ML models, and improving the logic for traffic signal pre-emption without causing the project's overall timeline to be delayed. This ability to react quickly to changing technical and simulation requirements made sure the system continued to meet performance and latency objectives. Eventually, while preserving development continuity and responsiveness to testing feedback, Agile principles made it possible to successfully integrate sophisticated features like multi-modal sensor fusion, dynamic traffic signal pre-emption, route prediction, and dashboard visualization.

2.1.1. Feasibility Study / Planning

The feasibility study examines the viability of designing and implementing a multi-modal emergency vehicle detection system integrated with the SUMO traffic simulator. This system encompasses features such as real-time visual detection from frontal and top-view cameras, OCR-based label extraction, acoustic siren recognition, and hue-based light bar detection to dynamically trigger traffic signal pre-emption. The present section provides a comprehensive analysis of the technical, operational, economic, and legal feasibility of the proposed solution.

1. Technical Feasibility:

- **Data Availability and Field Collection:** Following the initial project proposal and subsequent panel suggestions, the data collection strategy was significantly enhanced to include OCR-based top-angle detection, specifically designed to identify emergency vehicles caught in traffic blockages. To ensure high-fidelity, real-world accuracy, extensive primary data was gathered through targeted field visits. Fire engines were physically inspected at the Vavuniya Municipal Council, ambulances were examined at the Vavuniya Hospital and at SLIIT (in collaboration with the 1990 emergency service), and police vehicles were inspected at the Cheddikulam Police Station. This fieldwork allowed for the capture of distinct vehicle types, structural labels, visual profiles under varying lighting, and crucially, the specific acoustic differences between ambulance and fire engine sirens.

- **Hardware and Software Requirements:** The working prototype was developed and executed on a standard laptop capable of handling concurrent video and audio streams. Real-time visual and acoustic inputs were captured utilizing standard webcams, with the incoming data stored in a database and continuously retrieved for real-time processing. The software stack comprised YOLO, OpenCV, and Tesseract for visual and OCR processing, alongside YAMNet for acoustic siren detection. To physically demonstrate the system's traffic pre-emption capabilities, an Arduino board was integrated to dynamically visualize and control the traffic signal changes triggered by the detection algorithms, successfully bridging the gap between software processing and hardware actuation.
 - **Model Development:** The system utilizes a multi-modal fusion pipeline. YOLO processes the visual feeds, while HSV color segmentation and Haar Cascade classifiers are deployed specifically for light bar recognition. Simultaneously, a machine learning acoustic model processes audio feeds to isolate siren frequencies. The OCR module was specifically trained on the top-view angles captured during fieldwork to extract vehicle labels when frontal views are obstructed in a traffic jam.
2. **Economic Feasibility:** This study considered both development and operational costs. The total cost of the project was estimated at LKR [Insert your actual total cost], as detailed in Table 1.

Processing Unit	Jetson AGX Orin Dev Kit		LKR 450,000 – 600,000
Input Sensors	Cameras	Bullet Cameras (Front View)	LKR 75,000 – 135,000
		4K Dome / PTZ Cameras (Aerial View)	LKR 120,000 – 300,000
	Microphone Directional Shotgun Microphones (2 units)		LKR 30,000 – 90,000
Networking (VERY IMPORTANT)	PoE Switch (8-port) Ethernet cables + connectors		LKR 30,000 – 60,000

- **Resource Allocation:** The development team comprised researchers possessing specialized knowledge in computer vision, acoustic machine learning, traffic simulation (SUMO), and legal compliance. Responsibilities were allocated based on individual strengths, with specific members assigned to conduct the primary field research.
 - **Return on Investment (ROI):** The platform provides considerable return on investment by significantly reducing emergency response times (ERT), minimizing fuel wastage caused by traffic idling, and potentially saving lives and property by ensuring unobstructed paths for first responders.
3. **Legal and Ethical Feasibility:**
- **Legal Compliance (Sri Lankan Context):** A critical aspect of this study's feasibility involved validating the system's detection criteria against national laws to prevent false pre-emption. Consultation with a legal professional confirmed the parameters outlined in the Motor Traffic (Amendment) ACT No. 8 of 2009, Section 67. The system's acoustic and hue-based detection algorithms were strictly engineered to recognize only legally defined emergency signals. Specifically, the system was programmed to trigger pre-emption only when detecting:
 - *Acoustic limits:* Multi-tone horns or harsh, shrill alarms, which the Act states may only be used by the fire brigade, police/armed services, or an ambulance responding to an emergency call.
 - *Visual limits:* Specific light combinations mandated by the Act, including:
 - Red light for the Fire Brigade.
 - Red and blue lights for Police vehicles.
 - Red light for Ambulances.
 - Blinking blue or amber light for other Commissioner-General approved emergency vehicles.
 - **Data Privacy and Ethics:** The system utilizes simulated and public traffic camera feeds. The algorithms were designed strictly for vehicle and siren detection, with no capability to retain personally identifiable information (PII) of drivers or pedestrians.
4. **Operational Feasibility:**

- **Multi-Modal Processing:** The system proved highly capable of simultaneously analyzing multiple data streams. The addition of the panel-suggested top-view OCR module ensured that even if a vehicle is boxed in by civilian cars (obstructing the frontal view), the system can still read the "AMBULANCE" or "FIRE" labels from above.
 - **System Deployment and Integration:** The detection engine was successfully coupled with the SUMO traffic simulator via TraCI. For the purposes of this research, a centralized processing architecture was chosen to handle the YOLO, acoustic, and OCR pipelines before sending the pre-emption command to TraCI. The dashboard visualization was deployed to effectively relay this real-time data to end-users and relevant authorities.
5. **Time/Schedule Feasibility:**
- **Project Timeline:** A comprehensive timeline was created, including essential milestones such as the initial proposal, the panel review (which prompted the addition of the top-view OCR), the field data collection phase in Vavuniya/Cheddikulam, legal consultation, model training, SUMO integration, and final testing. The iterative Agile methodology provided room for adaptability when integrating the panel's feedback and the legally mandated light/sound constraints. Figure 7 illustrates the Schedule Management plan.

	2025						2026				
Task	July	August	September	October	November	December	January	February	March	April	May
Feasibility study											
Evaluate feasibility and background study											
Requirement gathering											
Equipment gathering and testing											
Background survey											
Literature review											
Requirement analysis											
Software requirement specification											
Functional, non-functional requirement											
Proposal presentation											
Project proposal report											
Data collection											
ML model creation											
Model training											
Model tuning											
Designing wireframes											
SUMO and Software integration											
Deployment & maintenance											
Progress presentation 1											
Research paper											
Progress presentation 2											
Testing											
Final presentation and viva											

Social and Cultural Feasibility:

- **Acceptance and Impact:** The tool addresses a critical urban pain point , traffic congestion delaying emergency services. By automating the clearing of intersections, the system reduces stress for both commuters and first responders.
- **Bias, Fairness, and Public Trust:** Because the system's triggering mechanism is legally bounded by the Motor Traffic Act No. 8 of 2009, it eliminates the risk of "favoritism" or false triggering for unauthorized vehicles using illegal multi-tone horns or inappropriate flashing lights. This strict adherence to the legal definitions of red, blue, and amber emergency lights ensures public trust and prevents unnecessary traffic disruptions.

Risk Management Plan

To anticipate and mitigate problems with team availability, supervisor input, and stakeholder involvement, a comprehensive risk management plan was created. The risk management strategy is displayed in Table 2 was main risk faced in the phase of Emergency Vehicle Detection Component.

Risk	Trigger	Owner	Response	Response Required
Camera failure / poor video quality	low resolution, hardware faults	EVD Component Member	Check camera connectivity Replace/upgrade cameras Use backup camera feeds	Backup cameras, Network tools
Incorrect Vehicle detection	Low lightning, bad training data	EVD Component Member	Improve dataset, Tired new model and fine tuning	Training dataset, GPU resources
OCR failure	GPU Loading with frame rate	EVD Component Member	OCR are wrappers, not the engine itself.	Install OCR executable file and included lightweight script that runs a command-line instruction to trigger
Siren sound not detected	Background noise, weak microphone input	EVD Component Member	Use noise filtering	Audio sensors, Signal processing tools
Flashlight detection failure	Daylight interference, weak flashing lights	EVD Component Member	Adjust brightness thresholds Add masking to detect the difference	Video processing tools

Communication Management Plan

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives Proactive communication is essential for project success.

Communication should be:

- **Adequate:** Providing information in the right format and content.
- **Specific:** Tailored to the targeted audience.
- **Sufficient:** Including all necessary details.
- **Concise:** Clear and to the point, avoiding unnecessary repetition.
- **Timely:** Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1. **Microsoft Teams:** For formal updates, documentation, and detailed communications.
2. **Documents (MS Word and PowerPoint):** For reports, presentations, and detailed project documentation.
3. **Phone Calls:** For quick discussions and urgent matters.
4. **Meetings:** Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.
5. **Chats (WhatsApp):** For informal, quick exchanges and real-time updates.

2.1.2. Requirement Gathering & Analysis

The requirement analysis phase relied heavily on stakeholder consultation and real-world domain analysis to extract both functional and non-functional system requirements. A pivotal shift in the project's scope occurred during the panel evaluation, which mandated a new functional requirement for top-angle OCR to handle traffic congestion scenarios. To translate this requirement into actionable, ground-truth data specifications, direct engagement with emergency service providers was necessary.

Site inspections were conducted at the Vavuniya Municipal Council, Vavuniya Hospital, SLIT (in collaboration with 1990), and the Cheddikulam Police Station. This empirical approach allowed the research team to extract precise technical requirements for the ML pipelines, including exact text placement for OCR parsing, non-functional requirements for handling dynamic lighting, and specific acoustic frequency ranges required to accurately classify different emergency vehicle sirens.

Data Collection

To achieve robust detection under varied conditions, a multi-source dataset acquisition strategy will be used:

- **Public Web Scraping:** A foundational visual dataset was established by sourcing a diverse array of emergency vehicle images (ambulances, police cars, and fire trucks) from public web engines. These images were specifically selected to represent various lighting, weather, and traffic conditions, and underwent strict filtering for quality, relevance, and resolution prior to preprocessing.
- **Open-Source Repositories:** To supplement the custom data, several pre-labeled datasets from Kaggle were utilized. These included high-resolution emergency vehicle image datasets with bounding box annotations, urban traffic object detection datasets (to provide a balanced mix of emergency and non-emergency vehicles), vehicle indicator light datasets (for route prediction), and the UrbanSound8K dataset (to isolate baseline siren audio for acoustic training).

- **Primary Field Recording:** To fine-tune the models for real-world application, authentic data was captured by deploying mobile and static cameras at local intersections to record actual emergency vehicle footage. Concurrently, directional microphones were used to record siren audio across varying levels of ambient urban noise, ensuring the acoustic training data accurately reflected real-world conditions.

Project Requirements

The project requirements define the expected functionality and operational qualities of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component. These requirements ensure that the system meets both performance goals and real-world applicability.

Functional requirements delineate the precise features and functionalities that a system must incorporate to achieve its designated traffic management and emergency response goals. These requirements served as a fundamental basis for the design, development, and validation processes of the detection system.

- **Stakeholder Input Approach:** The requirement identification process extended beyond theoretical research, relying heavily on primary stakeholder engagement and expert consultation:

- Crucial system enhancements were derived directly from the initial project panel review, which identified the specific need for top-angle OCR detection to handle vehicles trapped in traffic blockages.

- Operational and technical requirements were gathered through targeted field visits with first responders. Personnel at the Vavuniya Municipal Council, Vavuniya Hospital, the 1990 ambulance service, and the Cheddikulam Police Station provided insights into actual vehicle deployment, structural labeling, and the practical acoustic variations between different emergency sirens.

- Additionally, ongoing consultations were held with a legal professional who assisted the development team in comprehending the strict legal boundaries of emergency signal

usage, ensuring the system's triggering mechanisms were fully compliant with national regulations.

Figure 8 illustrates the details of the field visits and stakeholder consultations conducted to gather technical specifications, vehicle profiles, and legal constraints from emergency service personnel.

Figure 8: Field Visit and Stakeholder Consultation Details

In addition to the collection of functional requirements, this phase encompassed the manual compilation of a highly specific, multi-modal dataset designed for the training and validation of the detection models. The ground-truth data was exclusively sourced from direct physical observations and legal statutes during the field visits, which included:

- Visual captures of distinct vehicle types, structural labels, and top-angle profiles (for YOLO and OCR training)
- Acoustic recordings of specific siren frequencies under varying noise conditions (for acoustic model training)
- Visual documentation of specific light bar configurations (for HSV segmentation training)
- Legal text excerpts from the Motor Traffic (Amendment) ACT No. 8 of 2009, Sec 67 (for rule-based system validation)

All resources were verified on-site and against legal documentation to ensure complete adherence to real-world emergency vehicle configurations and Sri Lankan traffic laws, thereby guaranteeing that the system's detection criteria maintain operational accuracy and legal compliance. Figures 9, 10, and 11 illustrate examples of the captured real-world vehicle profiles, structural labels, and the specific light bar configurations documented during the field visits.

Functional Requirements

Identification of Key Functionalities:

- Visual Detection -Accurately detect emergency vehicles in real-time using a YOLO-based model and OpenCV video processing.
- Acoustic Detection -Identify ambulance sirens using DSP techniques and machine learning models even in noisy urban environments.
- Hue-Based Light Bar Detection -Detect emergency vehicle light bars using HSV colour segmentation and Haar Cascade classifiers.
- Multi-Modal Fusion -Combine visual, acoustic, and hue detection outputs to increase detection accuracy and reduce false positives.
- Indicator Detection -Detect turn indicators of emergency vehicles to predict movement routes.
- SUMO Integration -Communicate with SUMO via TraCI API to pre-emptively adjust traffic light phases for emergency vehicle passage.
- Route Sharing -Transfer predicted routes to neighbouring simulated intersections for coordinated signal adjustments.
- Dashboard Visualization -Display real-time detection events, simulation updates, and traffic light statuses on a web-based dashboard.
- Data Logging -Store detection events, simulation changes, and performance metrics for later analysis.
- Authority Notification -Send alerts to relevant traffic management authorities when emergency vehicles are detected.

Non-Functional Requirements

The system shall:

- Performance -Operate with a detection latency of less than 500 ms per frame for visual and acoustic detection combined.
- Accuracy -Achieve an overall multi-modal detection accuracy of at least 95% in simulation testing.
- Scalability -Support integration into larger Intelligent Transportation Systems (ITS) with minimal modifications.
- Reliability -Maintain consistent detection performance under varying lighting, weather, and noise conditions.
- Security -Protect stored and transmitted data through encryption and secure communication channels.
- Maintainability -Ensure modular code structure with proper documentation for easy updates and debugging.
- Usability -Provide an intuitive dashboard interface that can be operated without advanced technical knowledge.
- Compatibility -Be deployable on both local GPU-enabled systems and cloud-based environments.
- Resource Efficiency -Optimize computational resource usage to run on mid-range hardware without significant performance drops.

2.1.3. Designing

2.1.4. Implementation

2.1.5. Testing

2.1.6. Deployment & Maintenance

2.2. Development Methodology

2.2.1. Visual Detection Module

Description: Implement a YOLO-based real-time detection module integrated with OpenCV to identify emergency vehicles from live or simulated traffic camera feeds. The model will be fine-tuned with a dataset containing diverse lighting, weather, and traffic conditions.

Technologies & Tools:

- YOLOv8 / YOLOv5 (PyTorch) -Object detection framework for real-time identification.
- OpenCV -For video frame extraction, preprocessing, and drawing detection bounding boxes.
- Python -Core programming language for detection and integration.
- CUDA -For GPU acceleration during detection tasks.

2.2.2. Audio Detection Module

Description: Develop a siren detection system using Audio detection and machine learning to recognize ambulance sirens in noisy urban soundscapes. Audio input will be processed to extract features such as MFCC for model training.

Technologies & Tools:

- Librosa -For audio signal processing and feature extraction.

- TensorFlow / Keras -For building and training the siren classification model (CNN/LSTM).
- Scikit-learn -For preprocessing and baseline ML models.
- PyAudio -For capturing audio streams in real-time.

2.2.3. Hue-Based Light Bar Detection Module

Description: Implement HSV colour segmentation to detect characteristic emergency vehicle light bars, combined with Haar Cascade classifiers for improved detection reliability under varying conditions. Figure 1 shows the sound detection of siren using NN to detect it and give the single output.

Technologies & Tools:

- OpenCV -For HSV colour space segmentation and image filtering.
- Haar Cascade Classifiers -Pre-trained or custom-trained for light bar detection.
- Python -For scripting and integration with other modules.

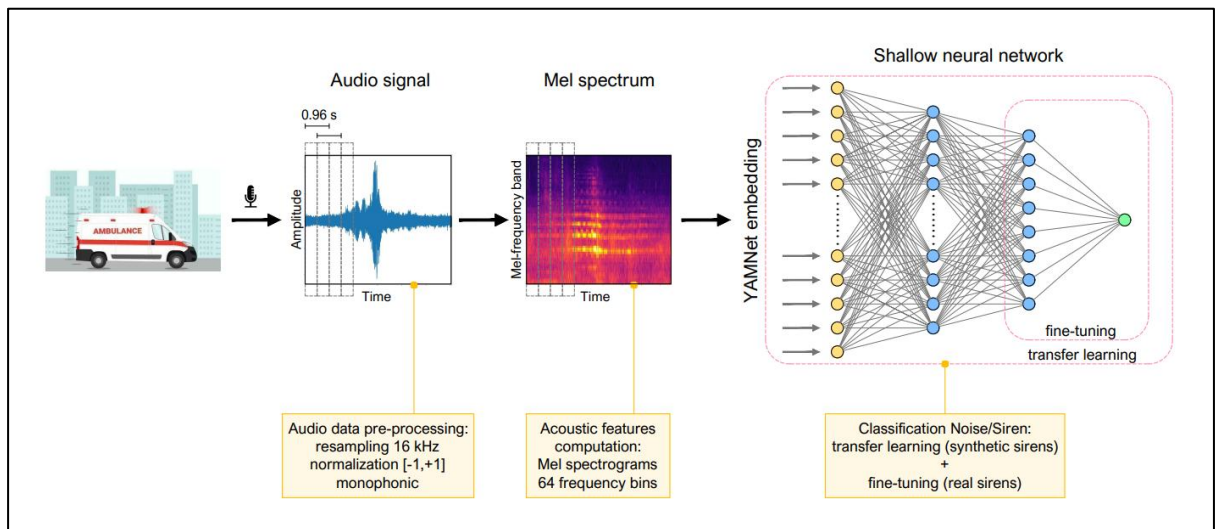


Figure 6 : Audio Siren Detection Source [14]

2.2.4. SUMO and TraCI Integration

Description: Connect the detection system to the SUMO traffic simulation environment using the TraCI API to trigger and adapt traffic light phases in real-time upon detecting an approaching emergency vehicle.

Technologies & Tools:

- SUMO (Simulation of Urban Mobility) -Traffic simulation software.
- TraCI (Traffic Control Interface) -Python API for controlling simulation events.
- NetEdit -SUMO network editor for creating and configuring intersections.

Figure 2 is the virtual environment where the traffic-based programs can be done it is more flexible and portable to use.



Figure 7 : SUMO Stimulator , Source : <https://networksimulationtools.com/sumo-simulator/>

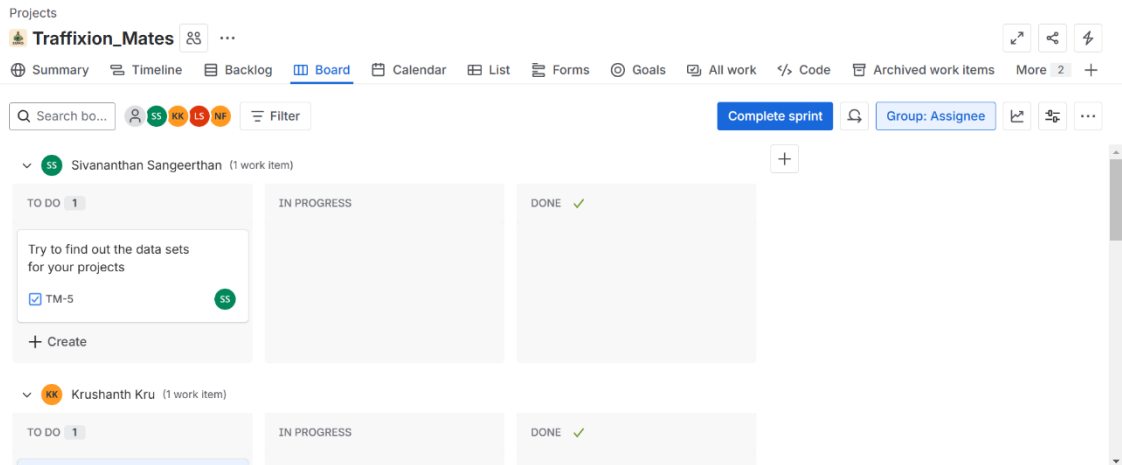


Figure 8 : Jira management tool

2.3. System Analysis

2.3.1. Software Solution Approach

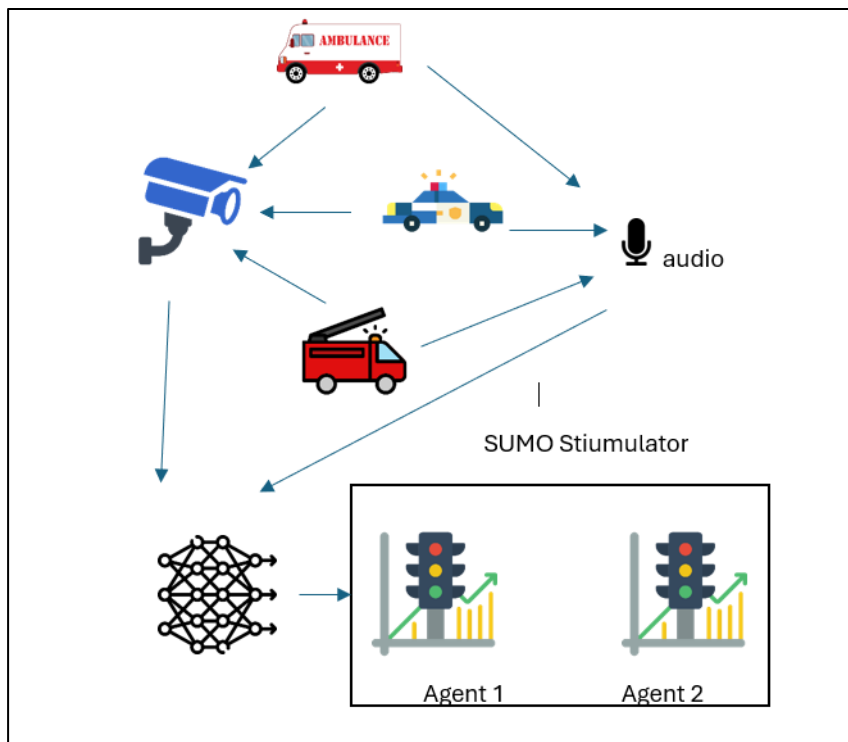


Figure 9 : System Diagram

3. RESULTS & DISCUSSION

4. FUTURE SCOPE

The Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component has strong potential for commercialization in both public sector and private sector markets. The system addresses a critical urban challenge reducing emergency response times making it valuable for deployment in Intelligent Transportation Systems (ITS) worldwide.

Potential Markets

- Government Traffic Management Authorities
 - City councils and transportation departments can integrate the system into existing traffic control centres to optimize emergency vehicle movement.
- Emergency Services
 - Ambulance, fire, and police departments can deploy the solution to ensure faster and safer passage through congested urban intersections.
- Smart City Solution Providers
 - Technology vendors offering IoT-based traffic management systems can incorporate the multi-modal detection module as a premium feature.

Future Commercial Potential

The technology can be extended beyond emergency vehicles to support:

- Public Transport Priority: Buses and trams receiving traffic signal priority during peak hours.
- Autonomous Vehicle Integration: Providing signal priority for driverless emergency and service vehicles.

5. CONCLUSION

The proposed Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component is a multi-modal intelligent traffic management solution designed to prioritize emergency vehicles in urban environments. By integrating YOLO-based visual detection and machine learning-based siren recognition, and HSV Cascade light bar detection, the system achieves high detection accuracy under varying conditions. Real-time integration with SUMO and TraCI enables dynamic traffic signal pre-emption, reducing response times and enhancing road safety.

The project follows an Agile development methodology, ensuring iterative improvement through testing phases including unit, integration, simulation, and performance testing. Expected outcomes include reduced travel times for emergency vehicles, improved intersection safety, and a scalable foundation for real-world deployment. With strong commercialization potential in government traffic control, emergency services, and smart city solutions, the system offers a cost-effective, high-impact contribution to modern urban mobility management.

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